

Task-2

Without using weights

Without Weights



Terminal ??? Task-2 [No Pretrained Weights]

```
$ python Task-2.py
```

```
[INFO] Loading model... (No Pretrained Weights)
```

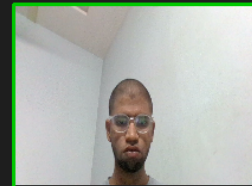
```
[INFO] Accessing webcam (VideoCapture(0))...
```

```
[INFO] Frame captured: 640x480
```

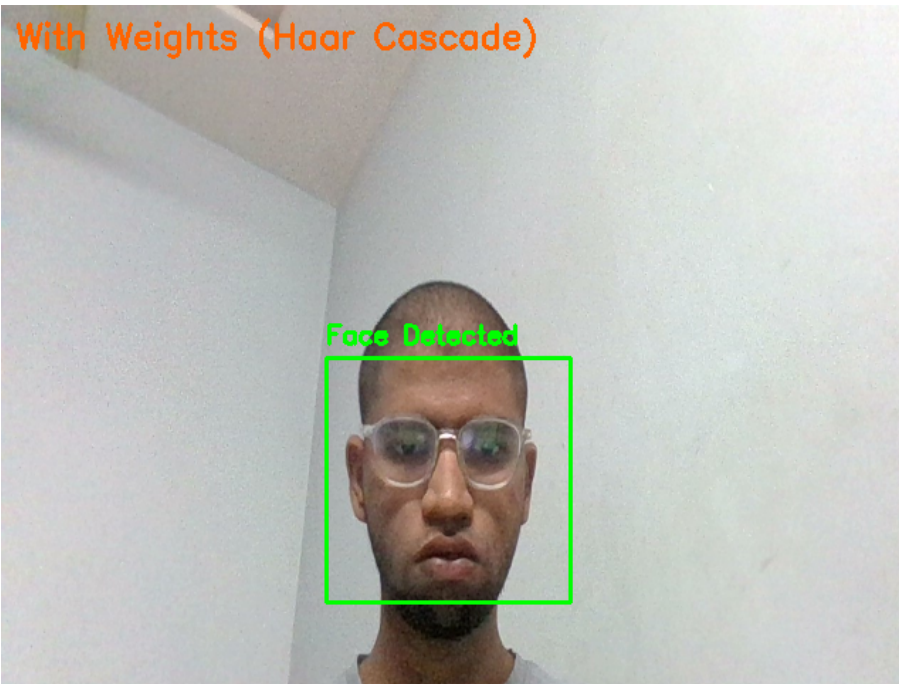
```
[INFO] Running inference on frame...
```

```
[INFO] Detection complete.
```

Live Frame



Using weights



Comparison: With vs Without Weights

If we are using weights it is slower but more accurate than without using the weights. Pretrained weights (e.g., Haar Cascade, ResNet, YOLO) allow the model to leverage learned feature representations, resulting in more reliable detections at the cost of additional computation during inference.

Terminal ??? Task-2 [Pretrained Weights (ResNet/YOLO)]

```
$ python Task-2.py
```

```
[INFO] Loading model... (Pretrained Weights (ResNet/YOLO))
```

```
[INFO] Accessing webcam (VideoCapture(0))...
```

```
[INFO] Frame captured: 640x480
```

```
[INFO] Running inference on frame...
```

```
[INFO] Detection complete.
```

Live Frame

